Brandon Fallin

Robotics and autonomy engineer specializing in distributed
multi-agent systems, learning-based control, and real-world robotic deployment.

+1 (727) 902 8910 | brandonfallin@gmail.com | brandonfallin | brandoncfallin.com

Education

University of Florida — **Gainesville, FL**
Doctor of Philosophy – Aerospace Engineering — *Aug. 2022 – May 2027 (Expected)*

University of Florida — **Gainesville, FL**
Master of Science – Aerospace Engineering, GPA: 4.00/4.00 — *Aug. 2022 – May 2024*

- Minor in Electrical Engineering, Certificate in Control Systems Theory
- Relevant Coursework: Nonlinear Control, Adaptive Control, Convex Optimization, Reinforcement Learning

University of Florida — **Gainesville, FL**
Bachelor of Science – Mechanical Engineering, GPA: 3.89/4.00 — *Aug. 2017 – May 2022*
Bachelor of Science – Aerospace Engineering

Technical Skills

Languages: Python, C++, MATLAB & Simulink, LaTeX
Robotics and Systems: ROS 2, Docker, Gazebo, PX4, Linux, Git, SolidWorks & PDM

Professional Experience

University of Florida Autonomy Park — **Gainesville, FL**
Robotics Team Lead — *Aug. 2023 – Present*

- Leads software integration, deployment, and maintenance for a fleet of 32+ autonomous air and ground vehicles including Freefly, Clearpath, Unitree, and custom PX4-based platforms
- Develops ROS 2 packages, Docker containers, and Gazebo simulation environments for sensor integration, localization, low-level control, and sim-to-real deployment across heterogeneous robotic platforms
- Managed a $120K+ AFOSR DURIP budget for procurement and integration of 20 custom autonomous quadcopters and ground-robot sensing upgrades
- Led 5 undergraduate researchers through hardware construction and conducted workshops on electronics, sensing, ROS 2, and PX4

Nonlinear Controls and Robotics (NCR) Lab — **Gainesville, FL**
PhD Candidate — *Aug. 2022 – Present*

- Researches distributed autonomy algorithms for heterogeneous multi-agent systems using nonlinear and adaptive control theory, sheaf theory, control barrier functions, and learning-based methods under the supervision of Dr. Warren Dixon
- Designed a distributed algorithm for online training of graph neural networks to approximate unknown drift dynamics with formal stability guarantees
- Implements and validates nonlinear control algorithms for multi-robot air/ground systems at UF Autonomy Park
- Coordinates with coauthors to publish theoretical and experimental findings in IEEE conference proceedings and journals

Air Force Research Laboratory — **Shalimar, FL**
AFRL Scholars Intern, Munitions Directorate — *May 2024 – Aug. 2024*

- Developed an online graph-learning algorithm combining GNNs, concurrent learning, and projected dynamical systems to infer unknown communication weights from multi-agent position measurements

Orbit — **Gainesville, FL**
Co-Founder — *Mar. 2021 – Aug. 2022*

- Closed a M&A deal with Creatd, Inc. (Nasdaq CM: CRTD)
- Managed development for a social media platform for retail investors on iOS through strategic planning and research for a customer-focused product

Professional Experience (Continued)

Power Design, Inc. **St. Petersburg, FL**
Mechanical Engineering Intern *Jun. 2021 – Aug. 2021*

- Worked alongside plumbing engineers and associate plumbing designers to produce technical drawings in Revit for multifamily high-rise, retail, and amenity spaces
- Produced technical unit plumbing plans for The Standard at South Carolina

Selected Publications

- **B. C. Fallin**, C. F. Nino, and W. E. Dixon, "GNN-based real-time graph learning using memory regressor extension," in Proc. IFAC World Cong., to appear.
- K. A. Currier, W. J. Leal, **B. C. Fallin**, J. Fairbanks, and W. E. Dixon, "From local to global: Sheaf-theoretic control barrier functions on manifolds," in Proc. IEEE Conf. Decis. Control, to appear.
- M. L. Gardenswartz, **B. C. Fallin**, C. F. Nino, and W. E. Dixon, "Collaborative indirect influencing and control on graphs using graph neural networks," in Proc. Am. Control Conf., 2026. [preprint]
- **B. C. Fallin**, C. F. Nino, O. S. Patil, Z. I. Bell, and W. E. Dixon, "Lyapunov-based graph neural networks for adaptive control of multi-agent systems," arXiv preprint, 2025. [preprint]
- **B. Fallin**, C. Hawkins, B. Chen, P. Gohari, A. Benvenuti, U. Topcu, and M. T. Hale, "Differential privacy for stochastic matrices using the matrix Dirichlet mechanism," in Proc. IEEE Conf. Decis. Control, 2023.

Experimental Validation

- C. F. Nino, M. L. Gardenswartz, **B. C. Fallin**, R. Kamalapurkar, and W. E. Dixon, "An exponentially convergent model-free differentiator," in preparation.
  - Wrote ROS 2 nodes for trajectory tracking and state estimation that enabled an aerial agent (Parrot Bebop 2) to maintain an exponentially convergent estimate of its velocity using a stream of position data from OptiTrack motion capture
  - Constructed a low-level PID control loop for the aerial agent to translate commanded velocity inputs into roll and pitch angle commands

- S. Akbari, X. Shen, W. Xue, J. C. Insinger, and W. E. Dixon, "Lyapunov-based adaptive transformer (LyAT) for control of stochastic nonlinear systems," in preparation. [preprint]
  - Conducted a trajectory-tracking experiment in which an aerial agent (Freefly Astro 2) was tasked with following a figure-8 trajectory while subjected to environmental disturbances. The aerial agent compensated for unknown drift dynamics using a transformer with a Lyapunov-based real-time weight update law

- J. W. Cross, "Experimental improvements to event-triggered indirect herding control of a cooperative agent," University of Florida, Undergraduate Honors Thesis, 2026.
  - Used a ROS 2 discovery server, OptiTrack motion capture, and a base station computer to conduct a 4-agent experiment in which an aerial shepherding agent (Parrot Bebop 2) used intermittent "pings" to guide 3 ground agents (ROSbot 2R Pro) to individual goal locations

- P. M. Amy, **B. C. Fallin**, J. N. Philor, and W. E. Dixon, "Event-triggered indirect control of a cooperative agent," IEEE Robot Autom. Lett., 2026. [pdf]
  - Implemented an event-triggered control law in ROS 2 for an aerial shepherding agent (Freefly Astro) that used intermittent "pings" to direct a target ground agent (Unitree Go1) to a goal location
  - Enforced operating region and inter-agent safety using agent-level potential field constraints